Mid Semester Review

Justin Le



SWARM EX COMMs

Original Plan and Milestones

- COMMs Lead for the semester
  - managing and tracking COMMs work
  - monitoring ground station bring-up
- Work towards Simulated Comms Test (SCT) 2 with Lithium 2 radio and Software Defined Radio in the loop
  - Re-running Minimum Communication Test
  - Re-running SCT-1
  - Running SCT-2
- Setting up Ground Station PC
  - implementing 2 machines, one Windows for Hydra, one Ubuntu 16 for GNU Radio
  - verifying the setup in the Ground Station Chain

Personal goals

- Develop in Hydra
- Develop in MPLAB for CDH
- Understand the packet chain end to end for the Lithium
- Embedded software skills
- Scope and track team work

How the tests build on each other

MCT
Hydra → CDH → Li-2 over UART

→ **SCT-1**
CDH ↔ Hydra hardwired

→ **SCT-2**
Adds the Li-2 and SDR, decoded through GNU Radio into Hydra

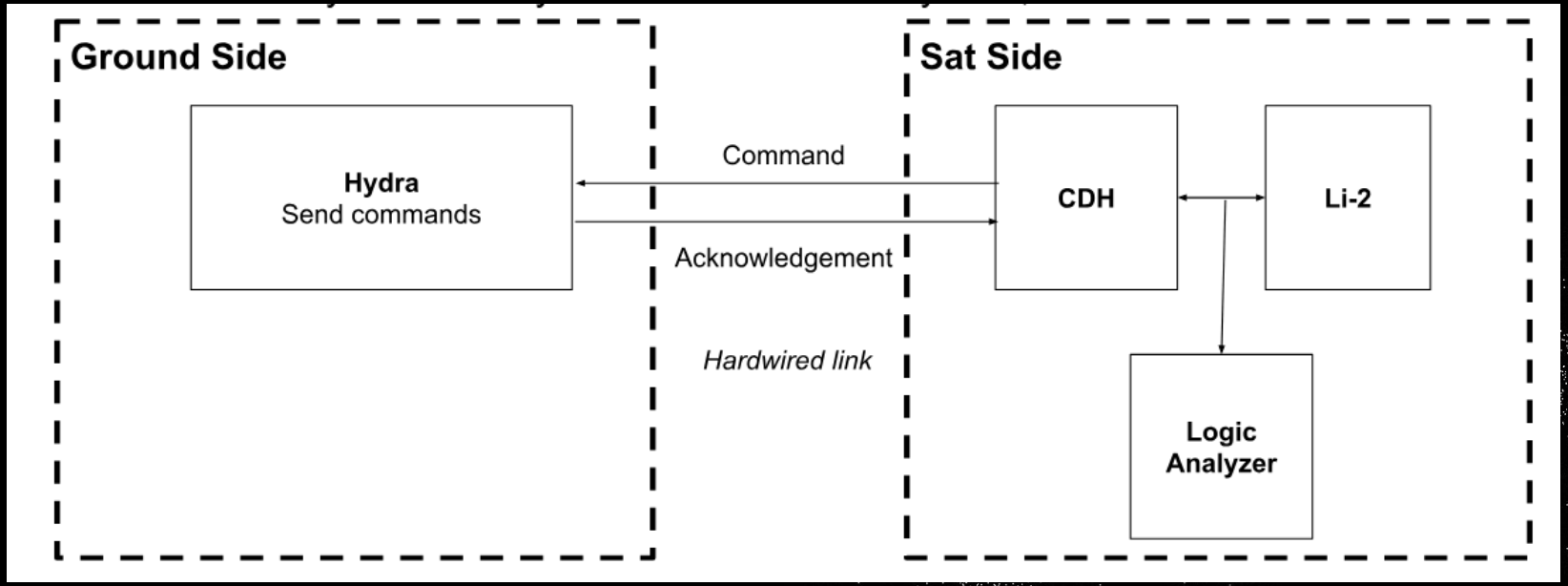
→ **SCT-3**
Adds the antennas, over a short range

→ **SCT-4**
Same setup, over a long range

Each test adds to the one before it, working toward CET and DITL. The next five slides show each test setup.

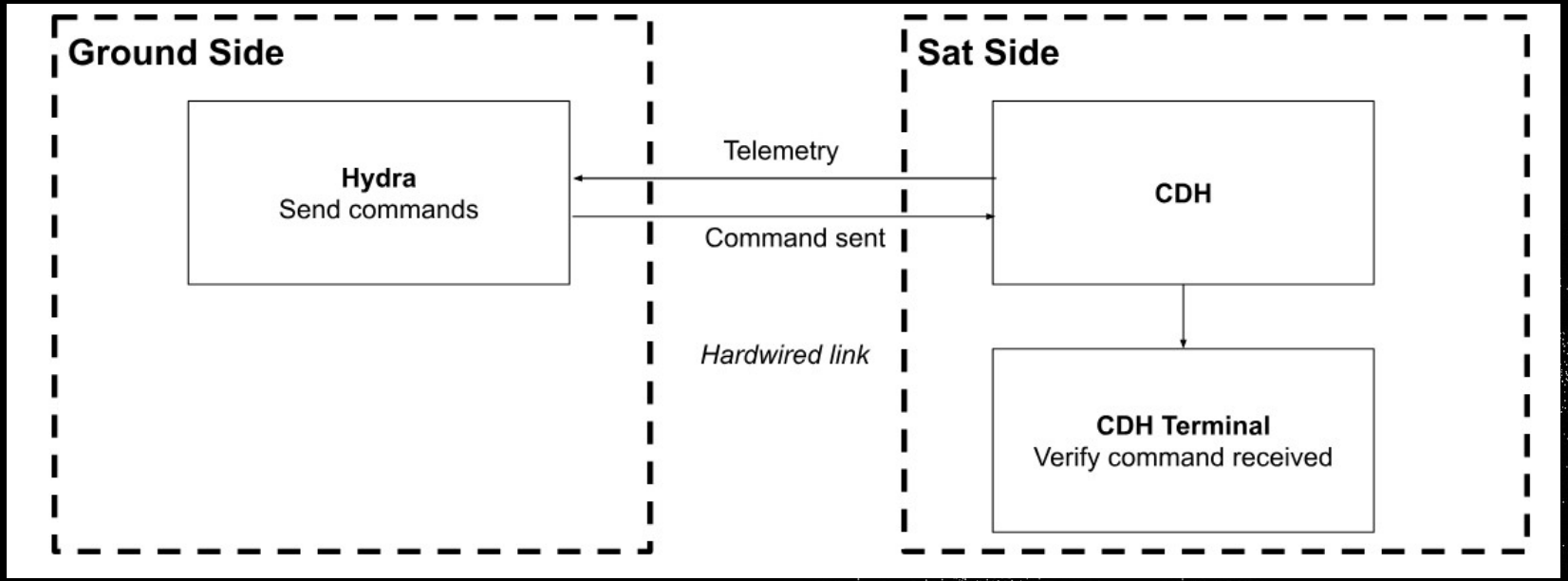
MCT: Hydra → CDH → Li-2 over UART

Verifies commands from Hydra reach the Li-2 through CDH and are acknowledged. A logic analyzer on the UART lines checks the no-op, transmit and reset commands.



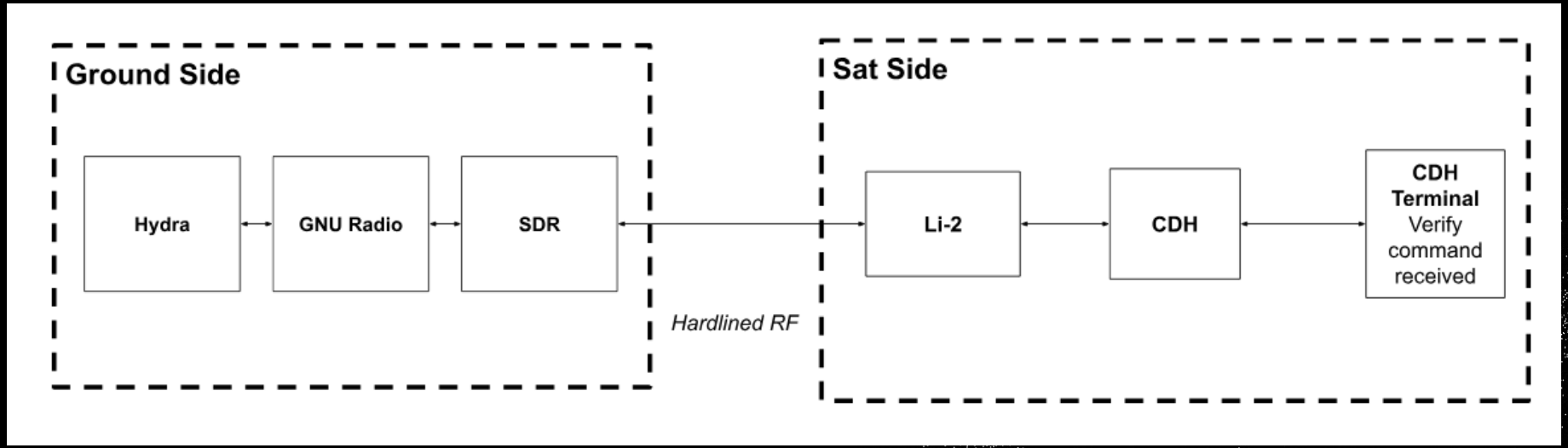
SCT-1: Hardwired CDH and Hydra

Verifies CDH reads Hydra's command bytes and Hydra reads CDH's telemetry bytes, with no wireless link. Hydra requests a housekeeping packet, the CDH terminal confirms the command arrived, and the packet shows up in Hydra telemetry.



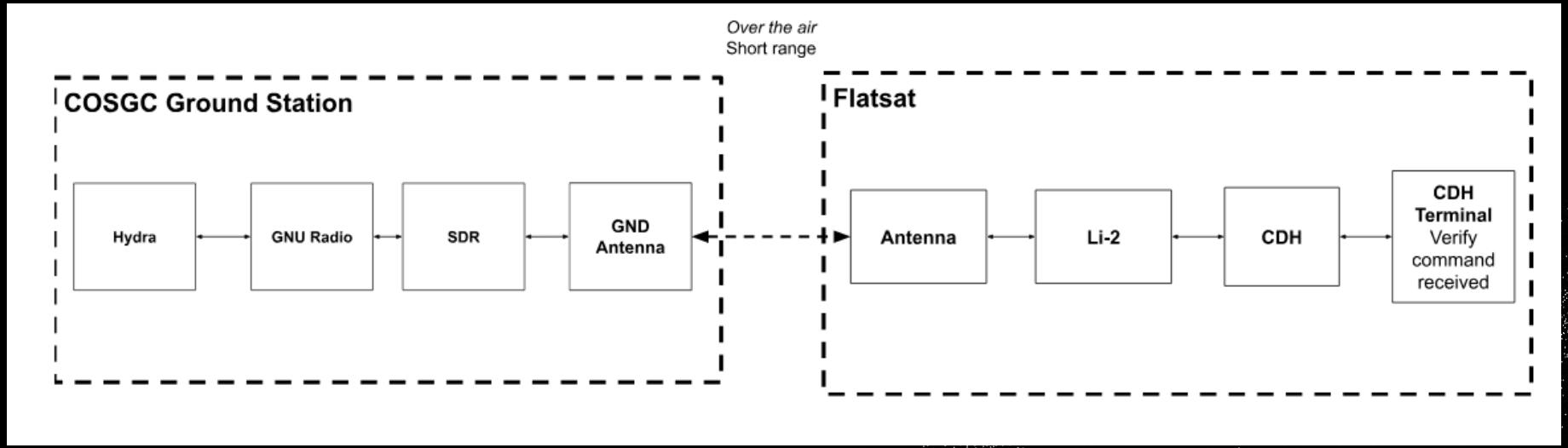
SCT-2: Full chain, antenna out of the loop

Verifies a message can go both ways through Hydra, GNU Radio, the SDR, the Li-2 and CDH. The SDR and Li-2 are connected by a hardline instead of antennas.



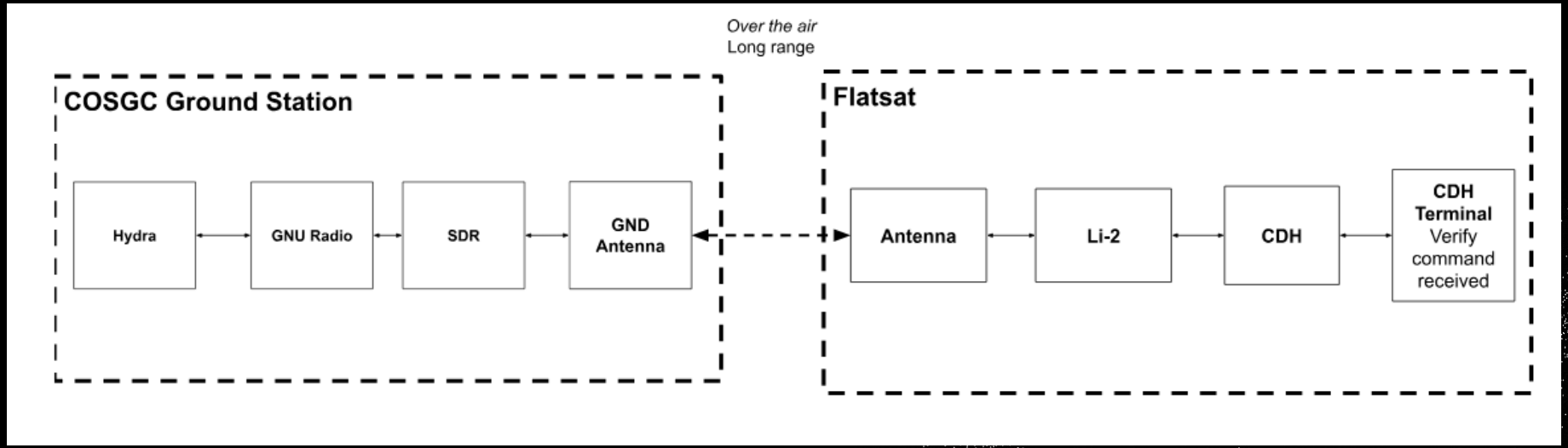
SCT-3: Antennas in the loop, short range

Verifies antenna and ground station performance and closing the link. The flatsat goes to the parking garage across from the DLC, and the SWARM-EX Hydra and GNU Radio instances run in the COSGC Ground Station.



SCT-4: Antennas in the loop, long range

Same setup as SCT-3. Only the distance between the flatsat and the ground station changes, to verify the link closes over a longer range.



Changes to the plan

Start of semester

- I was developing the GS PC architecture that will be implemented
- Rick's 2-PC proposal to mitigate hydra crashes
- The 443 white tower running Hydra, plus GS laptop running GNU Radio

Currently

- Scott Palo and Nicholas Rainville lead GS PC development and bring-up
- My role is verifying it in the comms chain once it works
- Blocked on development and no ground station licensing for a while

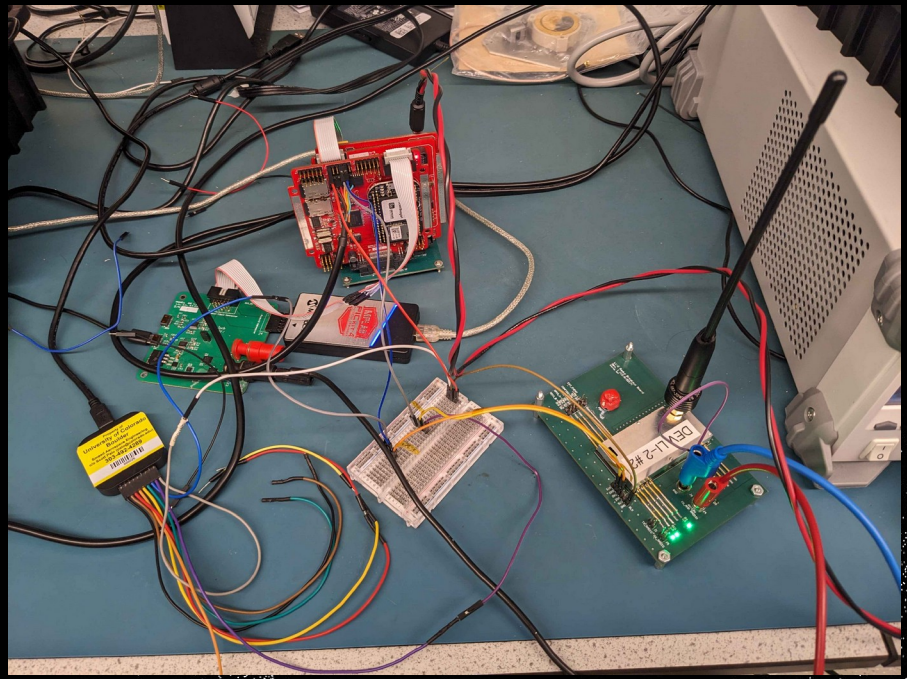
		Total Hours		Week Start Date																
		Estimated	Actual	8/24	8/31	9/7	9/14	9/21	9/28	10/5	10/12	10/19	10/26	11/2	11/9	11/16	11/23	11/30	12/7	12/14	Status
Technical	46%	129	43	0	2	3	7	8	15	8	0	0	0	0	0	0	0	0	0	0	
MCT: Hydra Lithium Communication		26	26			1	2	4	13	6											Late & Complete
MCT: Review test docs		1	4			1	1	2													Complete
MCT: Bench setup		2	3				1	2													Complete
Learning Hydra		4	2						2												Complete
MCT: Hydra development		4	4						4												Complete
MCT: Hydra hardwired to Li2		4	4						4												Complete
MCT: Hydra hardwired to CDH to Li2		4	2						2												Complete
MCT: Practice run-through of test		2	2						1	1											Late & Complete
MCT: Full run through of test		3	3							3											Complete
MCT: Troubleshooting		2	2							2											Complete
Ground Station PC Architecture		31	7		2	2	1		2												Not Started
Setting up laptop for GND station		8	1				1														Ontime (In Progress)
~~Connecting GND station with GND station se~~		5	0																		Not Started
Verify GS PC in comms chain		8	0																		Not Started
~~Planning final GND station architecture~~		10	6		2	2			2												Ontime (In Progress)

Minimum Communications Test

Worked bottom up, one interface at a time

1. Added the Li-2 commands, states and an opcode dictionary to Hydra
2. Hardwired Hydra straight to the Li-2 and confirmed no-op, transmit and reset on the logic analyzer
   - Took longer than planned from documentation gaps and getting familiar with the hardware and software
3. Put the CDH board in the loop. Commands were sent and acknowledged properly

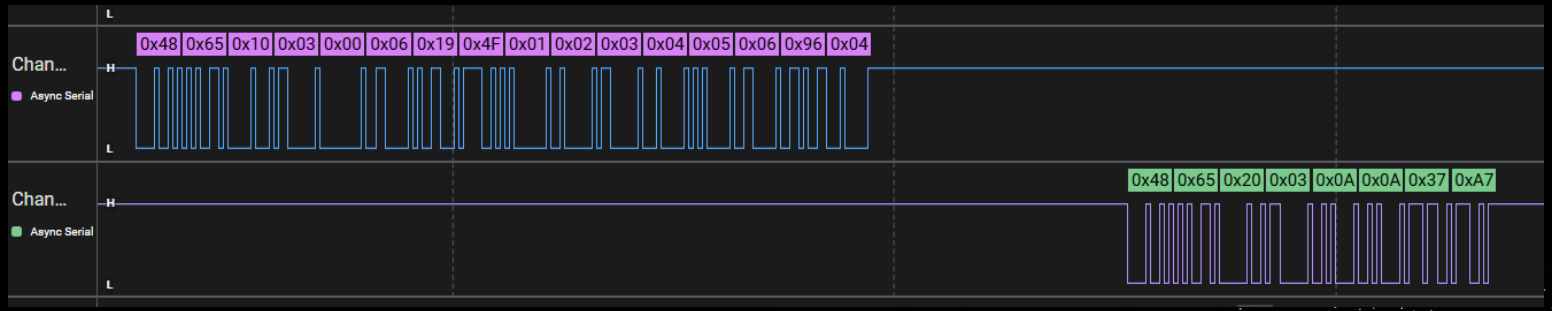
Result: commands acknowledged through Hydra → CDH → Li-2



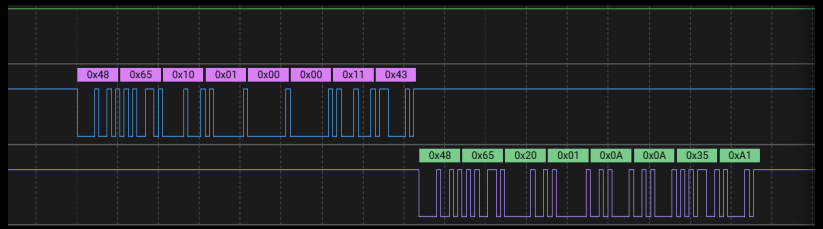
MCT: logic analyzer captures

Top trace: command sent to the Li-2. Bottom trace: the Li-2's reply

Transmit (0x03) with a payload



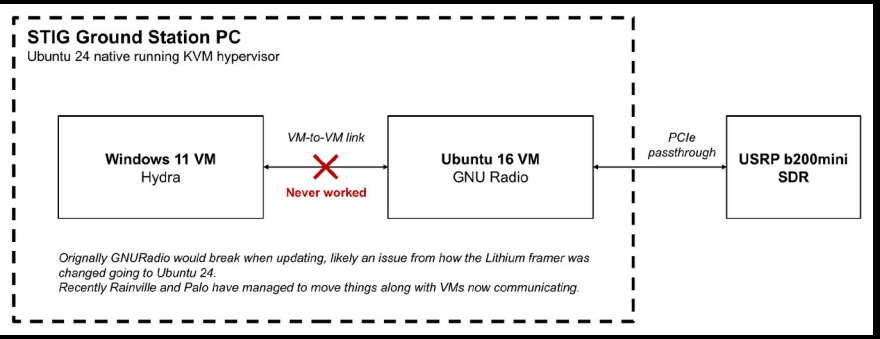
No-op (0x01)



Reset (0x02)



Ground Station PC



Created a STIG GS PC status doc

Covers the current setup, why it was built that way, the issues, who is involved, options and open questions. It became the shared picture for Palo, Rainville, and the MAXWELL and SWARM-EX COMMs teams.

Issues at start of the semester

- The VMs can't talk to each other
- VPN problems
- Our GNU Radio file can't run natively on Ubuntu 24
- There has been recent development and it can communicate with the SDR both ways

Verification

1. USRP → GNU Radio
2. GNU Radio ↔ Hydra
3. The whole ground side with the station hardware

Lead work

Team

- Onboarded Edin, Elsa, Atharv and Ian
- Elsa working on satellite tracking
- Edin working on ground station bring up
- Gave Atharv the SWARM-EX and ground station block diagrams
- Gave Ian the SCT documents for the SCT procedures

Tracking

- Tracking action items from the whole team along with ongoing issues
- Weekly quad charts for COMMs as a whole and a Gantt chart
- Managing blockers, dependencies, and rescheduling tasks for the team

Coordination

- Skye (MAXWELL)
- Alex Burns (former MAXWELL COMMs lead)
- Dhruva (former SWARM EX COMMs lead)
- Scott Gennari (LASP)

Schedule

SCT-1

10/16 – 10/22

Bench setup 10/16, practice run 10/19, full run 10/20–10/21, troubleshooting 10/22

GS PC verification

10/30 – 11/6

Once Palo and Rainville are ready to transfer to COSGC. Meanwhile, update GNU Radio and check its state for SCT-2

SCT-2

10/26 – 12/3

Procedure review 10/23, Hydra dev 10/27–11/2, practice 11/13, full run 11/17–11/20

Wrap-up

Weeks of 12/7 and 12/14

End-of-semester review, transition document and final report

Milestone	Planned	Now
Re-run MCT	9/21	Done 10/8
GS PC test	9/23	10/30 – 11/6
Re-run SCT-1	9/30	10/20
SCT-2	10/31	11/17 – 11/20

SWARM EX Impact

OnSat Comms

- Li-2 - CDH comms state was unclear
- Earlier semesters ran MCT and SCT but never reached SCT-2
- Re-verifying MCT and SCT gives a clear starting point for SCT-2

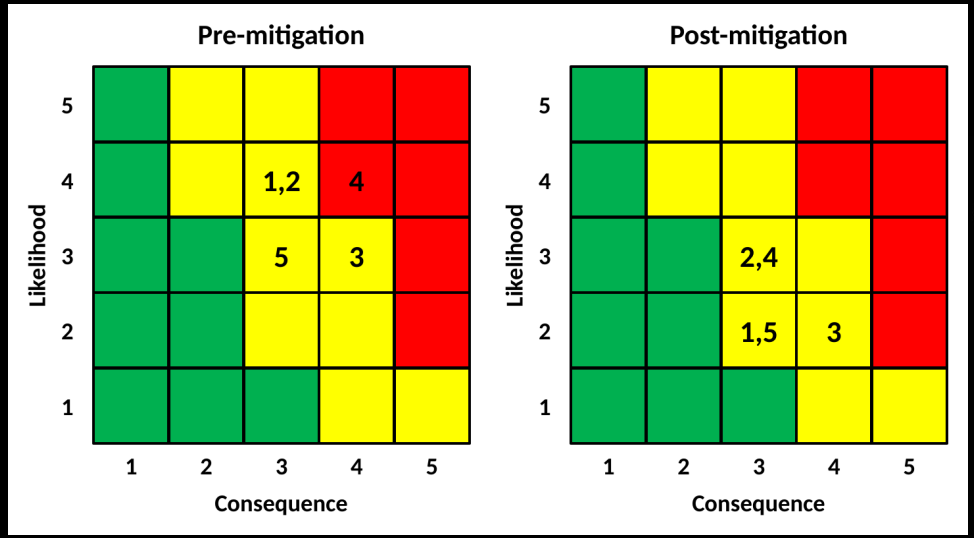
Ground station PC

- Long-running pain point for COMMs
- STIG doc turned it into an issue list that can be addressed
- Palo and Rainville bringing it up, needed for MAXWELL's February launch

COMMs lead

- Manage the tasks COMMs needs for SWARM-EX
- Team work tracked week to week
- Blockers addressed and routed to the right person (other subsystems, Palo, LASP, MAXWELL)

Risks and Concerns



1. **SCT-1 is delayed, which delays SCT-2**

   Mitigation: prep SCT-2 in parallel (procedure reviews, GNU Radio development) while SCT-1 finishes

2. **SCT-2 needs more development than planned**

   Mitigation: reaching out to Dhruva, Palo, or MAXWELL for support

3. **Motor interface can't be resolved or bypassed**

   Mitigation: Elsa is looking into working with Palo and getting help from LASP

4. **Delays in ground station bring-up**

   Mitigation: Allocating more people and time to focus more on the ground station

5. **Issues verifying the GS PC at COSGC**

   Mitigation: verify bottom up, with the interim laptop as a fallback

Thank you

